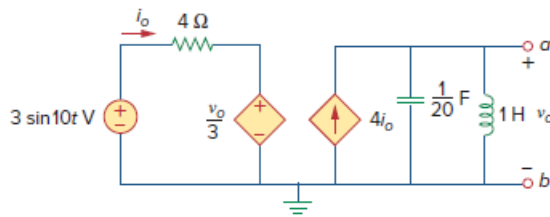


## Problem

Find the Thevenin equivalent at terminals  $a$ - $b$  in the circuit of Fig. 10.111.

Figure 10.111:



## Step-by-step solution

## Step 1 of 8

Refer to Figure 10.111 in the textbook.

The expression for voltage source is,

$$v = 3 \sin 10t \text{ V}$$

The phasor representation of current source is,

$$\mathbf{V} = 3 \angle 0^\circ \text{ V}$$

The angular frequency of the sinusoidal function is,

$$\omega = 10 \text{ rad/sec}$$

The value of capacitive reactance is,

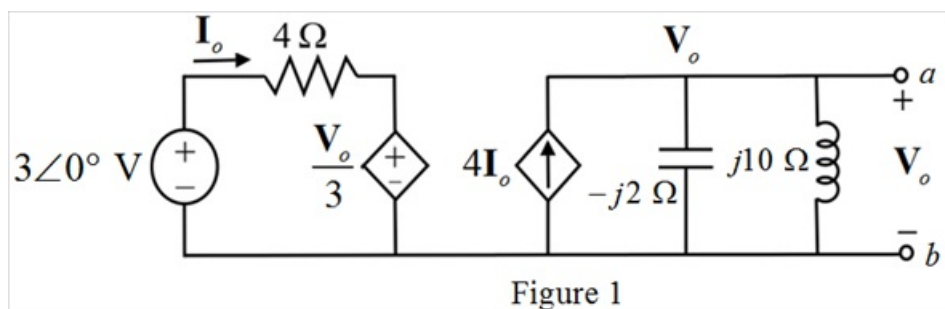
$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(10)\left(\frac{1}{20}\right)} \\ &= -j2 \, \Omega \end{aligned}$$

The value of inductive reactance is,

$$\begin{aligned} \mathbf{Z}_L &= j\omega L \\ &= j(10)(1) \\ &= j10 \, \Omega \end{aligned}$$

## Step 2 of 8

Draw the equivalent circuit diagram.



## Step 3 of 8

Determine the value of thevenin equivalent voltage.

The thevenin voltage is nothing but the open circuit voltage,  $V_o$  at the terminals a-b.

Therefore,  $V_{TH} = V_o$ .

Apply the Kirchhoff's Voltage Law to the input loop.

$$-3\angle 0^\circ + 4I_o + \frac{V_o}{3} = 0$$

$$4I_o = 3\angle 0^\circ - \frac{V_o}{3}$$

#### Step 4 of 8

Apply the nodal analysis at the node  $V_o$ .

$$-4I_o + \frac{V_o}{-j2} + \frac{V_o}{j10} = 0$$

$$\frac{V_o}{-j2} + \frac{V_o}{j10} = 4I_o$$

$$V_o \left( \frac{1}{-j2} + \frac{1}{j10} \right) = 4I_o$$

$$V_o (j0.4) = 4I_o$$

Substitute  $\left( 3\angle 0^\circ - \frac{V_o}{3} \right)$  for  $4I_o$  in the equation.

$$(j0.4)V_o = \left( 3\angle 0^\circ - \frac{V_o}{3} \right)$$

$$(j0.4)V_o = \frac{9\angle 0^\circ - V_o}{3}$$

$$(j0.4)(3)V_o = 9\angle 0^\circ - V_o$$

$$V_o (1.2j + 1) = 9\angle 0^\circ$$

$$V_o = \frac{9\angle 0^\circ}{1 + j1.2}$$

$$V_o = 5.76\angle -50.19^\circ \text{ V}$$

#### Step 5 of 8

The value of Thevenin equivalent voltage is,

$$\begin{aligned} V_{TH} &= V_o \\ &= 5.76\angle -50.19^\circ \text{ V} \end{aligned}$$

The time-domain representation of thevenin voltage is,

$$v_{TH} = 5.76 \sin(10t - 50.19^\circ) \text{ V}$$

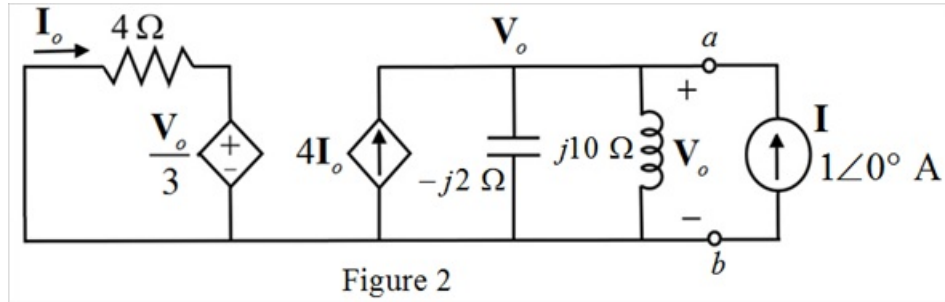
Thus, the Thevenin's voltage at the terminals a-b in the circuit is,

$$\boxed{5.76 \sin(10t - 50.19^\circ) \text{ V}}.$$

#### Step 6 of 8

Short circuit the voltage source and connect a test source  $1\angle 0^\circ \text{ A}$  across the terminals  $a$  and  $b$ , to determine the value of Thevenin's impedance,  $\mathbf{Z}_{\text{Th}}$ .

Draw the modified circuit diagram.



#### Step 7 of 8

Apply the Kirchhoff's Voltage Law to the left side loop.

$$4\mathbf{I}_o + \frac{\mathbf{V}_o}{3} = 0$$

$$4\mathbf{I}_o = -\left(\frac{\mathbf{V}_o}{3}\right)$$

Apply the nodal analysis at the node  $\mathbf{V}_o$ .

$$-4\mathbf{I}_o + \frac{\mathbf{V}_o}{-j2} + \frac{\mathbf{V}_o}{j10} - 1 = 0$$

$$-4\mathbf{I}_o + \mathbf{V}_o \left( \frac{1}{-j2} + \frac{1}{j10} \right) - 1 = 0$$

$$-4\mathbf{I}_o + \mathbf{V}_o (j0.4) - 1 = 0$$

Substitute  $-\left(\frac{\mathbf{V}_o}{3}\right)$  for  $4\mathbf{I}_o$  in the equation.

$$-\left(-\frac{\mathbf{V}_o}{3}\right) + \mathbf{V}_o (j0.4) - 1 = 0$$

$$\left(\frac{\mathbf{V}_o}{3}\right) + \mathbf{V}_o (j0.4) = 1$$

$$\left(\frac{1}{3} + j0.4\right) \mathbf{V}_o = 1$$

$$\mathbf{V}_o = \frac{1}{\left(\frac{1}{3} + j0.4\right)}$$

$$\mathbf{V}_o = (1.2295 - j1.4754) \text{ V}$$

#### Step 8 of 8

Determine the value of the Thevenin's impedance.

$$\begin{aligned} \mathbf{Z}_{\text{Th}} &= \frac{\mathbf{V}_o}{\mathbf{I}} \\ &= \frac{(1.2295 - j1.4754) \text{ V}}{1\angle 0^\circ \text{ A}} \\ &= (1.2295 - j1.4754) \Omega \end{aligned}$$

Thus, the value of Thevenin's impedance,  $\mathbf{Z}_{\text{Th}}$  is  $(1.2295 - j1.4754) \Omega$ .